# Inferring Negative Information in Logic Programming

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October 23, 2014

# Closed World Assumption

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# CWA

Suppose $P$ is a logic program and $A$ is a ground atom.

## CWA

Infer $\neg A$ if $P \not\models A$, i.e. if $P$ does not entail $\neg A$.

Note: Only applies to ground atoms $A$.

# Example Usage of CWA

## Example LP1

$$
\begin{array}{rcl}
p(X) & \leftarrow & q(X). \\
q(X) & \leftarrow & r(X, Y). \\
r(a, b) & \leftarrow & . \\
r(a, c) & \leftarrow & . \\
r(b, c) & \leftarrow . &
\end{array}
$$

Set of ground atoms entailed by $LP1$ is $\{r(a, b), r(a, c), r(b, c), q(a), q(b), p(a), p(b)\}$.

So using CWA, we have: $P \models^{cwa} \neg q(c)$, $P \models^{cwa} \neg p(c)$, and so forth.

# Negation as Failure

Suppose $P$ is a logic program and $A$ is a ground atom. NAF says we should infer $\neg A$ if there is no SLD refutation of the goal $\leftarrow A$ from $P$.

## Finite Failure Set

$A$ is in the *finite failure set* $F_P$ w.r.t. $P$ iff there exists an integer $i \geq 0$ such that $A \notin T_P \downarrow i$.

Thus, $A \in F_P$ iff $A \notin T_P \downarrow \omega$.

# Finitely Failed SLD Trees

## SLD Tree

An SLD tree for goal $G$ w.r.t. logic program $P$ is a tree such that:

1. Each node is labeled with a goal and the root is labeled with $G$;
2. If a node is labeled with goal $G_1$ and there exists a clause $C_1$ in $P$ such that $G_2$ is a resolvent of $G_1$ and $C_1$, then the node labeled $G_1$ has a child labeled $G_2$.

An SLD-tree is finitely failed iff the SLD-tree is finite and there are no branches ending in the empty clause.

# Theorem

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Suppose $P \cup \{G\}$ has a finitely failed SLD-tree of depth $k$ or less. Then so does $P \cup \{G\theta\}$ for any $\theta$.

**Proof sketch.** Suppose $P \cup \{G\}$ has a finitely failed tree of depth $k$. Let $p$ be an arbitrary path of length $r$ from root to leaf. $p$ has the form $G = G_0, \ldots, G_k$ and $G_k$ contains no atom that unifies with the head of any rule in $P$. Let $G_i$ $(i > 0)$ be obtained by resolving $G_{i-1}$ with a rule $C_i \in P$ via mgu $\theta_i$. Suppose we replace $G_0$ in this path by $G_0\theta$. Then one of two things occurs. The sequence $G_0\theta, G_1\theta, \ldots, G_k\theta$ is such that no atom in $G_k\theta$ unifies with any rule head in $P$ in which case we are done, or the sequence $G_0\theta, \ldots, G_k\theta$ ends abruptly prior to reaching the $k$'th resolution step. In which case finite failure occurs at or before level $k$.

# Theorem

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Suppose $P \cup \{\leftarrow A_1, \ldots, A_n\}$ has a finitely failed SLD-tree. Then there exists an $1 \leq i \leq n$ such that $P \cup \{\leftarrow A_i\}$ has a finitely failed SLD tree.

# Theorem

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Suppose $P \cup \{\leftarrow A\}$ has a finitely failed SLD-tree of depth $k$ or less. Then $A \notin T_P \downarrow k$.

**Proof sketch.** By induction on $k$.
Base Case If $P \cup \{\leftarrow A\}$ has a finitely failed SLD-tree of depth 1, then $A$ does not unify with the head of any rule in $P$ and so $A \notin T_P \downarrow 1$.

Inductive Case Suppose $P \cup \{\leftarrow A\}$ has a finitely failed SLD-tree of depth $(k+1)$. Consider an arbitrary child $G'$ of $A$ in the SLD-tree. Clearly at least one atom $A''$ in $G'$ has a finitely failed SLD-tree of depth $k$, so $A'' \notin T_P \downarrow k$ for all such atoms $A''$. As the children of $\leftarrow A$ in the SLD-tree are all possible resolvents of $P$ w.r.t. $\leftarrow A$, we know that for each rule in $P$ whose head unifies with $A$, the resulting goal is one of the children of $\leftarrow A$. For each of these children, one $A'' \notin T_P \downarrow k$, so $A \notin T_P \downarrow k+1$ as each rule whose head unifies with $A$ has a $A''$ in it which is not in $A'' \notin T_P \downarrow k$.

# Theorem

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Suppose $P$ is a logic program and $A \in B_P$. The following statements are equivalent:

1. $A \in F_P$;
2. $A \notin T_P \downarrow \omega$;
3. $A$ has a finitely failed SLD tree.

# Discussion

**Question**

Does $P \models^{cwa} A$ iff $P \models^{naf} A$?